Summary Factual Inconsistency Detection Based on LLMs Enhanced by Universal Information Extraction

Anonymous ACL submission

Abstract

Text summarization models have a potential flaw that causes the generation of summaries with factual inconsistencies. Recent studies leverage the remarkable inference capabilities of Large Language Models (LLMs) for factual inconsistency detection. However, compared with traditional metrics, LLMs remain unstable and inaccurate, and can lead to redundant inferences. To address these challenges, we propose a novel detection framework, UIEFID (Universal Information Extraction-enhanced Factual Inconsistency Detection). The idea is to define a self-adaptive UIE schema to guide LLMs in extracting structurally consistent semantic information from documents and summaries, ultimately detecting inconsistencies in the extraction results. The results show that the inference cost of UIEFID is only 24% to 31% of that of early LLM-based methods, enhancing efficiency and explainability.

1 Introduction

Text summarization aims to compress extensive source materials into brief summaries, including diverse content such as news articles, source codes, and cross-lingual text (Pu et al., 2023). Recently, Large Language Models (LLMs) demonstrate remarkable proficiency in addressing various downstream tasks (OpenAI, 2024), such as text summarization, dialogue systems, and machine translation.

LLM-based summarization integrates both extractive and abstractive methods (Figure 1). LLMs extract significant sentences information from the source document and then, under the constraint of brevity, compress and reorganize the extracted information. However, inherent hallucination (Zhang et al., 2023) in LLMs may result in the generation of summaries with factual inconsistencies, which hallucinate new information or generate content that contradicts the source document (Cao et al., 2017; Maynez et al., 2020). Obviously, factual inconsistencies can be introduced at any phase during the extraction and abstraction processes. In Extraction phase, models may generate inconsistencies in entities, attributes, relationships, actions, and emotions. In Abstraction phase, in pursuit of brevity, models may not only lead to loss and obfuscation of information but also potentially introduce additional knowledge to engage in over-inference. Pu et al., 2023 indicate a clear preference among human evaluators for LLMs-generated summaries over human-written summaries. It means that once LLMs generate factual inconsistencies, they can easily deceive human evaluators. Consequently, there is a critical demand for more robust methods to detect factual inconsistencies.

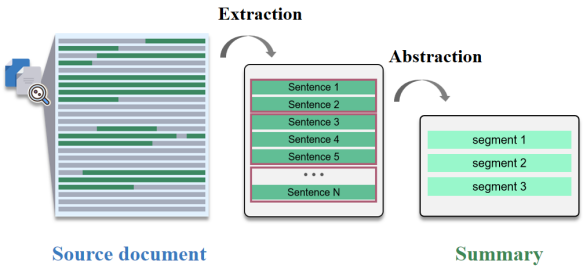


Figure 1: A simple summary generation process with two phase: Extraction and Abstraction.

Among existing inconsistency detection methods, employing LLMs as detectors has become a prevailing trend. There exist two main challenges in LLM-based methods: **(1) Efficiency.** Current methods heavily rely on Natural Language Inference (NLI) capability of LLMs. It means that LLMs need to perform extensive reasoning to analysis inconsistencies between summaries and source documents, whether these inconsistencies are obvious or implicit. Especially for long documents, LLMs are difficult to provide detection results within the expected time. **(2) Explainability.** LLM-based methods may themselves contain rea-

soning errors and produce unstable outputs. Such flaws render the detection results unreliable and deficient in explainability.

To address these challenges, we turn our attention to the Universal Information Extraction (UIE) capability of LLMs. UIE can rapidly process text, output structured information, and facilitate easier comparison and verification of the extracted information. It compensates for the shortcomings of NLI. Consequently, we present a novel detection framework UIEFID (Universal Information Extraction-enhanced Factual Inconsistency Detection). UIEFID obtains a fine-tuning LLM with excellent UIE capability, enabling it to extract corresponding structured information from both the summary and the source document. We perform detection on the structured information to identify the sources of inconsistencies. Then, we mark the sources and provide feedback to the LLM to conduct more refined reasoning.

Our proposed method, based on the characteristics of the two phases of text summarization, decouples UIE and NLI capabilities of LLMs. The structured information extracted by UIE is not only easy to compare and verify but also limits the scope of reasoning performed by NLI, enhancing the efficiency and explainability of the detection.

The contribution of this work can be summarized as threefold:

(1) Our study pioneers the integration of UIE to enhance the efficiency and explainability of LLM-based factual inconsistency detection.

(2) We propose UIEFID, a novel detetion framework that can rapidly perform and output structured detection results with strong explainability.

(3) The comprehensive experiments on popular AGGREFACT benchmark demonstrate the effectiveness of UIEFID, and the analysis for deeper understanding of the framework is provided.

2 Related Work

2.1 Factual Inconsistency Detection

Recently, research on factual inconsistency detection (also known as factuality evaluation) in text summarization has been highly active. The current detection metrics mainly fall into the following three paradigms.

NLI-based metrics. The Dependency Arc Entailment (Goyal and Durrett, 2020, DAE) leverage the entailment between the dependency parse trees of document sentences and the summary. SummaC-Cov (Laban et al., 2022) enables NLI models to be successfully used for this task by segmenting documents into sentence units and aggregating scores between pairs of sentences. Chen and Eger, 2023 proposed a series of NLI-based indicators for evaluating the factuality of summaries. They utilized the source document as the premise for the NLI system and the entire summary as the hypothesis for verification. AlignScore (Zha et al., 2023) provides a novel alignment system, which aligns the summary with large blocks of text from the source document and provides an overall score for entire summary.

QA-based metrics. This category of metrics typically depend on a Question Generation (QG) model designed to formulate questions based on the input document as context, and check whether information in the summary can be used to answer such questions, and vice versa. FEQA (Durmus et al., 2020), QAFactEval (Fabbri et al., 2022) and QuestEval (Scialom et al., 2021) are representative approachs among these metrics.

LLM-based metrics. Recent studies (Shen et al., 2023; Wang et al., 2023b) have shown that LLM-based metrics exhibit stronger evaluation performance compared to NLI-based metrics. TrueTeacher (Gekhman et al., 2023) analyzed the LLMs' capability of generating large-scale factuality datasets and then trained smaller student models on such data, outperforming existing NLI metrics. The recent trend of methods revolve around prompt engineering strategies, where LLMs are typically instructed with task description, the input document, and system-generated summary. Luo et al., 2023 were among the first to utilize LLMs for the detection of factual inconsistencies, inputting the source document in SUMMAC (Laban et al., 2022) along with its summary into GPT-3.5 for evaluztion. Fu et al., 2023 demonstrated the capabilities of LLMs in achieving multi-aspect, customized, and training-free evaluation. G-Eval (Liu et al., 2023) experimented with chain-of-thoughts (Wei et al., 2024a, COT) and form-filling instruction paradigms. However, early attempts have shown that LLMs underperform traditional models due to their limited ability to follow instructions and the absence of an effective detection methodology (Tang et al., 2023).

2.2 Universal Information Extraction

Information Extraction (IE) is a crucial domain in natural language processing that converts plain text into structured knowledge (Xu et al., 2023) . There has been a recent surge of interest in generative IE methods (Qi et al., 2023; Parvez et al., 2021) that adopt LLMs to generate structural information. Xu et al., 2023 categorize universal IE frameworks into two formats: natural language (NL-LLMs based) and code language (Code-LLMs based).

Lu et al., 2022 firstly propose a unified text-to-structure generation framework named UIE, which can universally model different IE tasks, adaptively generate targeted structures, and collaboratively learn general IE abilities from different knowledge sources. InstructUIE (Wang et al., 2023a) is a UIE framework based on instruction tuning in LLMs, which can uniformly model various information extraction tasks and capture the inter-task dependency. ChatIE (Wei et al., 2024b) is a two-stage framework to transform the zero-shot IE task into a multi-turn question-answering problem. Code4UIE (Parvez et al., 2021) is a universal retrieval-augmented code generation framework based on LLMs,which adopts Python classes to define task-specific schemas of various structural knowledge in a universal way.

3 Approach

3.1 Problem Definition

Given a non-empty document $D = \{d_i\}_{1 \le i \le N}$ and a corresponding summary $S = \{s_j\}_{1 \le j \le M}$, where D contains N sentences and S contains M sentences, typically $N \ge M$, and the text length of S does not exceed that of D. This indicates that converting D into S involves a mapping from multiple sentences to a single sentence, or a mapping from a single sentence to a single sentence.

$$f(\{d_p, \cdots, d_q\}_{1 \le p \le q \le N}) \rightarrow \{s_j\}_{1 \le i \le M} \quad (1)$$

Inspired by text compression algorithms (Li et al., 2022) , we present a concise hypothesis: applying a lossless structured compressor to D and S, transforming the semantics and logic of the textual expression into Unified Structured Information (USI). This hypothesis can be formalized and defined as follows:

$$compressor(D, S) \rightarrow (I_d, I_s) \quad (2)$$

where $compressor(\cdot)$ is a function that converts document D and summary S into their corresponding structured information, I_d and I_s, respectively.

The compression results not only completely reflect the semantic entities, logical relationships, and positional markers in text but also significantly reduce the substantial redundancy of content that needs to be processed, accelerating machine detection. In Figure 2, the summary is factually consistent with the document, and the lossless structured information from the document must entail the structured information of the summary theoretically.

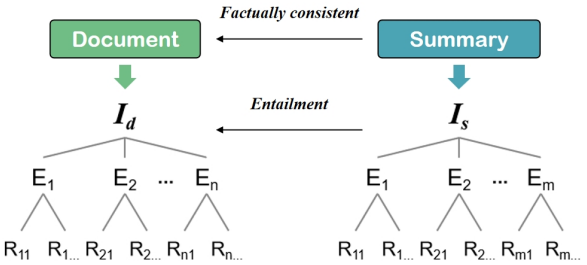


Figure 2: Lossless text compression (tree structure). $E_{\cdot}$ represents an entity, and $R_{\cdot\cdot}$ represents the relationship corresponding to each entity.

In grammar, the subject is the entity who performs the action or is described. To express I_d and I_s concretely, we transform I_d and I_s into a subject-centered triplet format, described as follows:

$$I_d, I_s \rightarrow \{Sbj_i : (k_j, v_j)_{1 \le j \le k_i}\}_{1 \le i \le n/m} \quad (3)$$

where Sbj_i denotes the i-th subject in D or S, k_j, v_j respectively denote the j-th key-value pair belonging to Sbj_i, which express specific attributes or relations, n and m respectively represent the total number of subjects present in D and S, and k_i indicates the number of key-value pairs possessed by the i-th subject.

Additionally, we need to define the following evaluation function to quantify the consistency as:

$$FID = \begin{cases} 1 & \text{if } mScore(I_d, I_s) \ge t; \\ 0 & \text{otherwise,} \end{cases} \quad (4)$$

where FID is a label for factual inconsistency detection, which is determined by calculating the matching score $mScore$ between I_d and I_s. If $mScore$ exceeds the threshold t, the output is 1 (consistent); otherwise, the output is 0 (inconsistent).

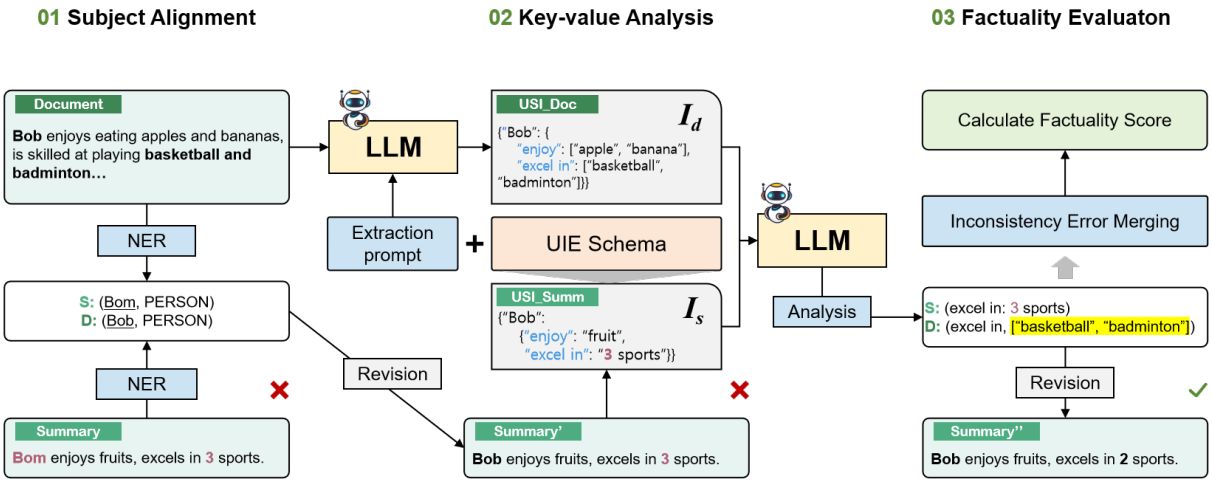


Figure 3: UIEFID framework consists of 3 phases: **(i) Subject Alignment.** Identifying and correcting inconsistencies in subjects between the document and the summary. **(ii) Key-value Analysis.** Extracting USI from the summary and using it as a extraction schema to guide the fine-tuning LLM to extract USI from the document, and then prompting LLM to reason and analyze the differences in structured key-value pairs and obtain the detection results. **(iii) Factuality Evaluation.** Collecting inconsistencies from two phases and merging them to calculate the final factuality score.

3.2 UIEFID

To concretize the detection idea mentioned above, we propose a novel detection framework **UIEFID (Universal Information Extraction-enhanced Factual Inconsistency Detection).**

Unlike previous detection methods, we abandon the use of sentence-level validation and instead adopt a more efficient structured information validation. We perform structured compression mapping of source documents and summaries through universal information extraction, and then, in the structured information, prompts LLMs to conduct semantic comparisons and locate the corresponding sentences or paragraphs in the source document where inconsistent statements are found. (see Figure 3)

UIEFID adopts an iterative strategy that combines detection and revision for external and internal inconsistency, divided into 3 phases according to the order of execution: ***Subject Alignment***, ***Key-value Analysis*** and ***Factuality Evaluation***.

3.2.1 Subject Alignment

A summary typically consists of statements involving semantic elements such as subject, action, object, modification, and state assertions. In this paper, the term *"subject"* is used to encompass all terms related to named entities, which is equivalent to the intermediate nodes of a tree-structured information (see Figure 2).

In this phase, we compare the subject differences in the structured information by Named Entity Recognition (NER) , and identify three types of errors: *non-existent*, *spelling errors*, and *misplacement*. Specifically, ***non-existent*** indicates that the subject in the summary does not have a corresponding subject in the source document; ***spelling errors*** means that the subject in the summary was not correctly output; ***misplacement*** refers to the relative position of the subject in the summary being different from that in the source document.

Algorithm 1 describes the subject alignment process. After identifying inconsistencies, we perform text similarity computation and coreference resolution to select appropriate subjects from the source document to replace the inconsistent content. Meanwhile, the analysis results of subject alignment are recorded as a basis for subsequent calculations of factuality scores.

Clearly, if the subject extracted from the summary is significantly different from that in the source document, it will misguide the LLMs and introduce bias in the further extraction of structured information. To prevent over-reasoning when directly using LLMs to detect factual inconsistencies, we believe it is necessary to promptly correct subject errors. This ensures that LLMs can solely focus on more complex logical inconsistencies during subsequent detection phases.

Algorithm 1 Subject Alignment

Input: doc, $summ$
Output: $inconsist_{ents}$

$unmatched_{ents} = null$
$doc_{ents} = NER(doc)$
$summ_{ents} = NER(summ)$
for $ent_i \in summ_{ents}$ **do**
  if $ent_i \notin doc_{ents}$ **then**
    $unmatched_{ents}.append(ent_i)$
  end if
end for
for $ent_i \in unmatched_{ents}$ **do**
  $entsList = doc_{ents}[ent_i.label]$
  $maxScore = MaxSim(ent_i, entsList)$
  $ent_{maxsim} = MaxSim^{-1}(maxScore)$
  if $maxScore \leq threshold$ **then**
    $inconsist_{ents}.append(ent)$
  else
    Replace ent_i in $summ$ with ent_{maxsim}.
  end if
end for
return $inconsist_{ents}$

3.2.2 Key-value Analysis

On the basis of eliminating subject inconsistencies, we utilize LLMs to extract structured information from the summary in the form of triplets (s, p, o), and s, p and o represent the subject, predicate, and object, respectively.

In this phase, we define the self-adaptive schema based on the triplets extracted from the summary, which is used as a prompt schema to guide LLMs in universal information extraction.

Most LLM-based methods tend to use the document as a reference to detection for inconsistencies in the summary, a method that can be easily affected by the length of the document, leading to a decrease in performance. In contrast, our method uses the summary as the provider of the UIE schema, making the scope of inconsistencies to be detected mainly determined by the length of the summary. Since the summary is usually much shorter than the document, our method can transform the natural language inference task on long texts into a more localized and specific UIE problem, thereby improving efficiency and accuracy.

Figure 5 illustrates the key-value analysis process. The initial step involves subject alignment in document-summary pair to ensure subject consistency. For instance, "Rovers" or "Kevin" are not mentioned in the document, and these non-existent subjects are removed. Next, the revised summary is input into a fine-tuning LLM to extract USI. During this process, the detailed value of each key-value pair is masked by the marker "(?)", forming the UIE schema. Subsequently, the schema and document are input together into the model, guiding it to extract and complete corresponding detailed values based on entity relationship constraints. Finally, by comparing the two structured information extracted from the summary and document, we identify an inconsistency in the "contract" entry: the summary states that the contract period between Featherstone and Duffy is 2 years, while the document shows it is only for 1 year.

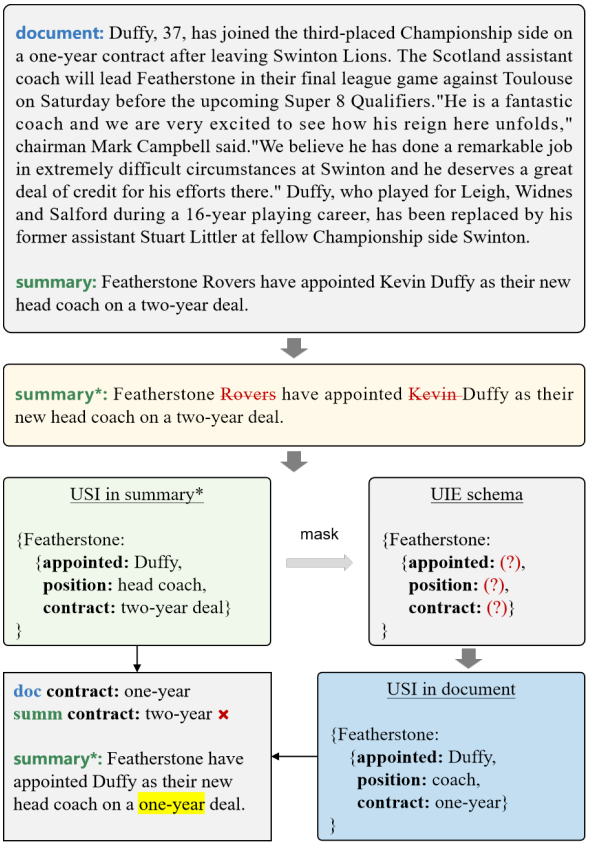


Figure 4: An example of Key-value Analysis based on Unified Structured Information.

3.2.3 Factuality Evalution

UIEFID maximizes the use of LLMs' universal information extraction (UIE) capabilities, significantly boosting the efficiency and interpretability of the detection process. However, to quantitatively assess the factuality of summaries within a vast array of text summarization tasks, a robust scoring system remains essential. After completing the detection and correction, we quantify the accumu-

lated inconsistencies to obtain a summary factuality score. We establishe the formula for factuality score (FS) using the aforementioned symbols, as follows:

$$FS = w_1 \times \frac{|ents_{fc}|}{|ents|} + w_2 \times \frac{\sum_{i=1}^{|sbj|}(\frac{|kv_{fc}|}{|kv_{sbj_i}|})}{|sbj|} \quad (5)$$

where $ents_{fc}$ represents entities consistent with the source document, and kv_{fc} represents key-value pairs (reflecting relationships or events) that are consistent with the source document.

Since different types of errors exist in the two phases, we assign different weight coefficients to each phase, where w_1 represents the coefficient for the subject alignment phase, and w_2 represents the coefficient for the key-value analysis phase.

Factuality score essentially measures the degree of consistency between the summary and the source document. Using the source document as a reference, we calculate the ratio of consistent entities present in the summary, as well as the ratio of key-value pair matches in unified structured information obtained after universal information extraction.

4 Experiment and Results

4.1 Experimental setting

We introduce experimental settings, including the benchmark, evaluation metric, and baselines.

Benchmark. We evaluate the effectiveness on AGGREFACT, the established benchmark for assessing summarization factuality metrics. AGGREFACT aggregates 9 existing annotated summary factuality datasets of news articles. All datasets contain summaries generated from articles in CNN/DM and XSum. Given the unique characteristics of CNN/DM and XSum, our proposed benchmark includes two subsets, AGGREFACT-CNN and AGGREFACT-XSUM, that evaluate the performance of factuality metrics on these two datasets separately. Tang et al., 2023 stratify it according to the underlying summarization model, categorized into FTSOTA, EXFORMER and OLD based on their development timeline. (1) FTSOTA represents state-of-the-art fine-tuned summarization models, including BART (Lewis et al., 2020), PEGASUS (Zhang et al., 2020), and T5 (Raffel et al., 2020). (2) EXFORMER is a collection of early Transformer-based summarization models, comprising BERTSum (Liu and Lapata, 2019) and GPT-2 (Radford et al., 2019). (3) OLD covers the remaining models, such as Pointer-Generator (See et al., 2017) and BottomUp (Gehrmann et al., 2018).

Evaluation Metric. We use balanced accuracy to evaluate the performance of factual inconsistency detection methods due to the imbalance of factually consistent and inconsistent summaries in AGGREFACT dataset.

Baselines. We evaluate the effectiveness of UIEFID against 12 recent baselines mentioned in (Scirè et al., 2024). These baselines are categorized categorized into 3 types based on their detection pattern, including NLI, QA, and LLM-based metrics (see Section 2.1) .

4.2 Implementation Details

For each document-summary pair to be examined, we use spaCy's en-core-web-trf as an entity extractor to identify the entities contained in both the document and the summary. Subsequently, we filter out potential misaligned entities by performing character matching and similarity calculation on the entities in both the document and the summary. Misaligned entities serve as a basis for judging inconsistency. And we utilize a coreference resolution model Coreference to select the true nouns or phrases that are closest to the misaligned entities from the source document to replace them.

Moreover, we select the latest LlaMA3-8B as the base model and trained a fine-tuned model capable of universal information extraction (Wei et al., 2024b). We prompt the model to extract unified structured information from the revised summary, which comprises entities, relationships and events, and then to mask the detailed values of the extracted semantic structure to create UIE schema in form of JSON. Further, we require the model to extract and complete the masked values based on the source document.

4.3 Results and Analysis

We divide UIEFID into two distinct variants based on whether LLM-based extractor is used in every phase.

UIEFID$_{base}$ utilizes external NLP tools for processing during the subject alignment phase and only employs a fine-tuned LlaMA3 model for assessing factual consistency in summaries during the key-value analysis phase. It takes into account

that there are already mature and low-resource dedicated models for the entity recognition task.

UIEFID$_{llm}$ uses LLM-based prompt in both the subject alignment and key-value analysis phases. It considers the simplicity and fluidity of the processing workflow, opting to use LlaMA3 to complete all the phases.

Table 1 presents the balanced accuracy scores on the AGGREFACT test sets. UIEFID$_{base}$ sets a new state-of-the-art performance on AGGREFACT, resulting in the highest average balanced accuracy score, with the alternative variant UIEFID$_{llm}$, ranking second.

Table 1: Balanced acccuracy results on the test sets of the AGGREFACT-CNN and AGGREFACT-XSUM datasets (threshold-per-dataset setting). A trivial baseline that predicts all examples as factually (in) consistent reports a balanced accuracy of 50%. Following Tang et al., 2023, we exclude the performance of DAE on the EXFORMER (EXF in table) and OLD datasets in the AGGREFACT-XSUM partition, since it was trained on XSumFaith (Goyal and Durrett, 2021) which is part of those splits. Results in **bold** indicate the best performance, while underlined values represent the second best.

	Agg-CNN			Agg-XSum			
	FTS	**ExF**	**OLD**	**FTS**	**ExF**	**OLD**	**AVG**
Baseline	50.0	50.0	50.0	50.0	50.0	50.0	50.0
DAE*	59.4	67.9	69.7	73.1	-	-	67.5
QuestEval	63.7	64.3	65.2	61.6	60.1	59.7	63.5
SummaC-ZS	63.3	76.5	76.3	56.1	51.4	53.3	65.3
SummaC-Conv	70.3	69.8	78.9	67.0	64.6	67.5	70.5
TrueTeacher-11B	62.0	67.5	80.5	75.9	68.4	52.8	67.9
QAFactEval	61.6	69.1	80.3	59.0	59.6	60.5	67.0
MENLI	63.4	54.9	66.8	59.0	59.7	70.5	62.4
AlignScore	62.7	73.2	78.0	69.4	**77.5**	63.7	70.8
ChatGPT-ZS	66.2	64.5	74.3	62.6	69.2	60.1	55.7
ChatGPT-CoT	49.7	60.4	66.7	56.0	60.9	50.1	57.9
ChatGPT-DA	48.0	63.6	71.0	53.6	65.6	61.5	60.1
ChatGPT-Star	55.8	65.8	71.2	57.7	70.6	53.8	63.8
UIEFID$_{llm}$	71.3	72.8	82.2	71.2	70.5	80.2	74.4
UIEFID$_{base}$	**75.6**	**78.4**	**82.6**	**76.1**	74.5	**80.7**	**78.2**

UIEFID$_{base}$ and UIEFID$_{llm}$ have both achieved better results on AGGREFACT than previous models. Following the principle from extraction to abstraction, we divide reasonably the factual inconsistency detection process into two phases, which can adopt LLaMA-8B to handle finer details of errors. The experimental results also show that UIEFID performs better on average on CNN than on XSum, which is related to our design. Universal information extraction is more suitable for extracting a large amount of structured information from long texts. The summaries of XSum are generally shorter and more abstractive, making it difficult to explicitly extract specific relationships or events, which is a disadvantage affecting its performance.

4.4 Ablation study

We conducted an analysis of the main impacts of UIEFID to verify the necessity of leveraging the universal extraction capabilities of large models and to highlight the improved role of subject alignment (SA) and key-value analysis (KVA). We use UIEFID$_{ZS}$ as the baseline, utilizing only its inherent NLI capability to detect factual inconsistency without introducing any phase involved in our proposed phases. Next, we incrementally add subject alignment and key-value analysis to observe the impact of these two modules on the baseline's performance. Furthermore, we introduce the complete UIEFID process to the baseline, incorporating both the subject alignment and key-value analysis modules.

Table 2: The results of the following baselines: i) UIEFID$_{ZS}$, using only LLM's NLI capabilities in a zero-shot prompt, ii) UIEFID$_{SA}$, which performes subject alignment, iii) UIEFID$_{KVA}$, which performes key-value analysis, and iv) UIEFID$_{SA+KVA}$, integrating subject alignment and key-value analysis.

Metric	CNN	XSUM	AVG	ΔAVG
UIEFID$_{ZS}$	56.3±0.8	60.4±0.3	58.3	-
UIEFID$_{SA}$	58.7±1.7	64.8±0.5	61.8	+3.5
UIEFID$_{KVA}$	63.2±2.1	67.2±0.2	65.2	+4.6
UIEFID$_{SA+KVA}$	72.6±2.4	80.7±0.6	76.7	+11.5

Table 2 displays the balanced accuracy scores of UIEFID on the AGGREFACT FTSOTA split, comparing them to the aforementioned baselines. The values in the ΔAVG column represent the differences in terms of average balanced accuracy resulting from the incremental inclusion of each component. The results indicate that each component significantly contributes to the final performance. The alignment strategy with input text at multiple granularity has the most substantial impact, resulting in a significant gain of 11.5 points when included.

4.5 Performance

Simple Configuration. We eliminate the complex configurations of previous detection models, such as batch text data cleaning and claim extraction, and simply use readily available NLP tools and open-source LLMs. This will greatly reduce the barrier to using UIEFID.

Hallucination Control. LLMs rely on their reasoning capabilities to perform detection, which may exist hallucinations. Drawing on the definition of information entropy, the more precise the scope of detection, the higher the detection accuracy. We refine the focus of detection and filter out abundant document information with universal information extraction and self-adaptive schemas, and then guide LLMs to only focus on the factual detection of the specified parts.

Explainability. UIEFID effectively integrates qualitative and quantitative analysis strategies. Qualitative analysis reflects the factual consistency of the summary by the output of the differences in structured semantics between the summary and the document. These differences can be directly detected by specific syntax parsers. Quantitative analysis utilizes factuality score to calculate the comprehensive evaluation results of the subject alignment and key-value analysis phases.

Rapid Inference. LLM-based detection methods leverage their reasoning chain capabilities for detailed detection and explanation, and ensure the effectiveness of factuality detection. Consequently, this results in a bottleneck where the inference cost for detection increases proportionally with the length of the document, making it difficult to adapt to long text detection. Instead, we only require LLMs to extract structured information from documents that meets the schema of structured semantics reflected by summaries. On average, our shift to a summary-centric method does't cause excessive invalid inference.

Figure 5 illustrates that the average inference cost of UIEFID on AGGREFACT is significantly lower than other methods. The result confirms the improvement in detection efficiency brought by integrating universal information extraction.

Iterative Revision. Our proposed detection method provides a specific revision module for each phase to restore the partial factuality of the summary until it is completely consistent with the document. We attempt to ingeniously combine factuality detection with summary revision to enhance its application potential.

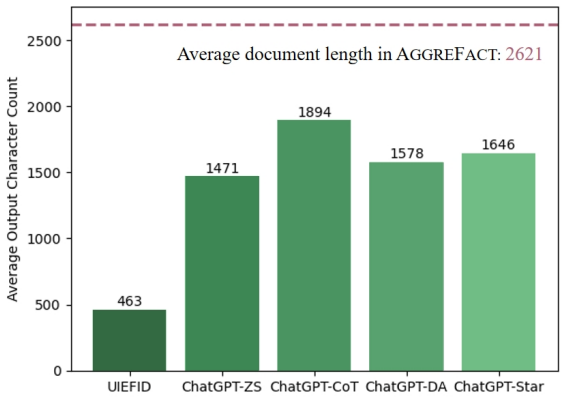


Figure 5: Average inference cost used by UIEFID is only 24% to 31% of that used by other LLM-based methods during the detection process.

5 Conclusion

Text summarization models have a potential flaw that causes the generation of summaries with factual inconsistencies. Inspired by structured semantic extraction and lossless text compression, we employ a LLM-based method, innovatively combine with Universal Information Extraction (UIE), and propose a novel UIEFID framework. It effectively enhances efficiency and explainability of summary factual inconsistency detection. We conduct comprehensive experiments to demonstrate the effectiveness of UIEFID on popular AGGREFACT benchmark, and also provide deeper analysis for ablation study. Our experiment results indicate that fully leveraging the remarkable universal information extraction capabilities of LLMs is a promising path to improve detection performance. UIEFID is capable of detecting factual inconsistencies in summaries and incrementally refining it until it is consistent with source documents. In addition, we evaluate the average inference cost used by UIEFID during the detection process, which is only 24% to 31% of that used by other LLM-based methods. Future research will consider adding a "self-reflection" module to LLM-based text summarization, where LLMs actively detect and provide feedback to correct the generated results through frameworks similar to UIEFID.

Limitations

Our proposed UIEFID detection method relies on the universal information extraction capabilities of LLMs. Currently, our findings further highlight the improvement in efficiency and explainability brought about by leveraging universal information extraction (UIE), but it doesn't provide conclusive evidence that the structured triplet representation is necessarily the best. Perhaps there is a better representation for expressing the structured semantic information of text. Moreover, we lack summary datasets generated by the latest LLMs, and have not conducted comprehensive analysis on the characteristics of LLM-based text summarization.

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